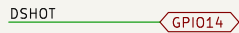
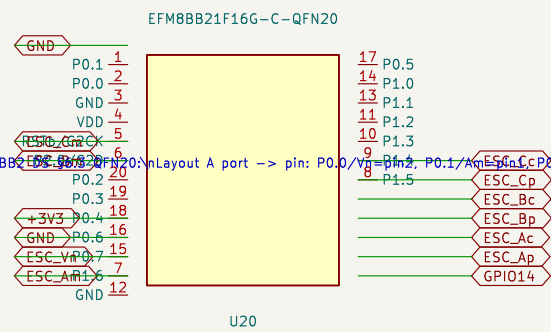


FLAG/GUI: place bee-core:EFM8BB21F16G-C-QFN20 + 3x A03401A + 3x A03400A + gate R +\nsense R network + VBAT pad/polyfuse + bulk C. Prefer correct nets over fake completeness.\nUNCERTAIN: P-FET full enhancement at Vgs=-(4.2-3.3)= -0.9V on 1S is NOT enough \n\whoop designs often gate P-FETs from VBAT-referenced drivers or use different FETs.\nRe-check OpenAI0-Whoop / BLHeli hardware before PCB. Consi



FET bridge (per phase, Layout A: Ap/Bp/Cp = PWM/high-side, Ac/Bc/Cc = COM/low-side):\nPhase A: Q_PA A03401A (P, C5296722) src=VBAT drain=PHASE_A gate=Ap(P1.0)\n Q_NA A03400A (N, C20917) src=GND drain=PHASE_A gate=Ac(P1.1)\nPhase B: Q_PB/Q_NB same with Bp/Bc (P1.2/P1.3)\nPhase C: Q_PC/Q_NC same with Cp/Cc (P1.4/P1.5)\nBEMF: Am/Bm/Cm (P0.1/0.2/0.3) sense via resistor divider to PHASE_A



U1 EFM8BB21 (bee-core lib) - physical pins from EFM8BB21: nLayout A port -> pin: P0.0/Pin=pin2, P0.1/Am=pin3, P0.2/Bm=pin20, P0.3/Cm=pin19\nP0.5/RTX=pin17, P1.0/Ap=pin18\nP1.1/Bp=pin13, P1.2/Bp=pin11, P1.3/Bc=pin10\nP1.4/Cp=pin9, P1.5/Cc=pin8. VDD=pin4 (+3V3), GND=pin3+12+EP, RSTb=pin5 (pull-up)\nP2.0/C2D=pin6 (debug). Bypass 100n+4.7u on VDD.

SECTION: ESC – EFM8BB21 + Layout A FET bridge (Bluejay)\nCited: Bluejay Layouts/A.inc – PWM_ACTIVE_HIGH, COM_ACTIVE_HIGH, PWM side HIGH.\nMCU: EFM8BB21F16G–C–QFN20 LCSC C80713. Silk: BLUEJAY LAYOUT A / VBAT MAX 1S.\nDShot: GPIO14 -- Rseries 100–330R --> EFM8 P0.5 (pin 17).\n1S direct–drive P+N whoop topology (no FD6288). Verify Vgs: 3.3V M

Nets via global labels to Minimal RP2350 GPIOs
Bee Core bring-up – graft sheet

Sheet: /
File: esc.kicad_sch

Title: Bee Core – ESC EFM8 Layout A

Size: A3

Date: 2026-09-15

Rev: v0.1-wip

KiCad E.D.A. 9.0.2

Id: ?/1